Project Initialization and Planning Phase

Date	01 July 2026
Team ID	xxxxx
Project Title	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to build an intelligent flood prediction system using machine learning, improving early-warning accuracy and response time. It tackles the gap left by conventional forecasting methods, promising better disaster preparedness, reduced risk to life and property, and more efficient resource allocation. Key features include a multi-model ML classification system and a Flask-based prediction interface.

Project Overview	
Objective	To build a machine learning-based flood prediction system that analyzes historical rainfall and weather data to accurately forecast flood risk and support timely disaster response.
Scope	The project covers data collection, exploratory analysis, preprocessing, training and comparison of multiple classification models, and deployment of the best-performing model as a Flask web application, with potential scalable deployment on IBM Cloud.
Problem Statement	
Description	Conventional flood forecasting methods often fail to predict floods with sufficient lead time, leaving authorities and communities with insufficient opportunity to respond, resulting in loss of life and property.
Impact	Solving this issue will enable earlier flood warnings, better evacuation planning, more efficient disaster resource allocation, and a reduction in flood-related damage and casualties.
Proposed Solution	
Approach	Employing multiple classification algorithms - Decision Tree, Random Forest, K-Nearest Neighbours (KNN), and XGBoost - trained on historical rainfall and weather data to predict flood likelihood, and selecting the best-performing model for deployment.
Key Features	- Implementation of a machine learning-based flood risk classification model. - Real-time flood risk prediction via a Flask web application. - Comparison of multiple algorithms to select the highest-accuracy model. - Scalable deployment designed for IBM Cloud.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU specifications	Intel i3 or above

Memory	RAM specifications	Minimum 4 GB
Storage	Disk space for data, models, and logs	Minimum 10 GB free space
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, XGBoost, Joblib/Pickle
Development Environment	IDE	Anaconda Navigator, Jupyter Notebook
Data		
Data	Source, format	Kaggle flood prediction dataset, csv